W. Dewmini Kawshani

Software Engineer Intern

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Summary

IT and Management undergraduate with hands-on full-stack development experience spanning e-commerce, health-tech, and IoT projects, including AI integration. Strong problem-solver and collaborative team contributor, quick to pick up new tools and technologies, seeking a Software Engineering Internship to apply and grow these skills on real-world projects.

Education

BACHELOR OF SCIENCE HONOURS IN INFORMATION TECHNOLOGY AND MANAGEMENT 03/2024 – 07/2028
University of Moratuwa
CGPA – 2.86 (First 3 semesters)

DIPLOMA IN INFORMATION TECHNOLOGY AND COMMUNICATION 03/2023 – 09/2023
ICBT campus

DIPLOMA IN ENGLISH AND LITERATURE 03/2023 – 02/2024
Aquinas Higher College of Studies

GCE ADVANCED LEVEL PHYSICAL SCIENCE STREAM 2023
Buddhist Girls' National College, Wennappuwa
Achieved BCC
Z-Score 0.9779

Technical Skills

Programming Languages	JavaScript, TypeScript, Java, C, Python
Frontend Development	React, TailwindCSS, HTML5, CSS3, Responsive Web Design
Backend Development	Node.js, Express.js, RESTful API Design, Server-Side Rendering
Databases	Relational: PostgreSQL, MySQL, MSSQL NoSQL: MongoDB, Firebase
Cloud & DevOps	Docker, Git, GitHub
Soft Skills	Team Collaboration, Agile, Problem Solving, Time Management, Communication

Projects

OneShop – E-Commerce & POS Platform *Team Project / Second-Year Software Project*

In this project, I focused on the online customer experience, covering authentication, customer account management, and order fulfillment. I designed the backend data models and built REST APIs for registration, login, profile management, checkout, and order tracking, secured with JWT authentication and a dual-path password recovery flow. I developed the customer-facing interfaces, including checkout with real-time delivery fee calculation by Sri Lankan district and a public order tracking page, and implemented an order lifecycle with stock reservation at each stage. I also wrote integration tests with Jest and Supertest to verify authentication and role-based access.

Technologies: Next.js, TypeScript, Tailwind CSS frontend, Node.js/Express.js backend, MongoDB with Mongoose, JWT, Nodemailer, Jest and Supertest.

Demo: opendoor.allinoneshop.store **GitHub:** ecommerce_web_app

LottoScan – Sri Lankan Lottery Ticket Checking Platform *Personal Software Project / Ongoing*

In this project, I am building a full-stack web application for checking Sri Lankan lottery results (NLB/DLB), covering all 16 national lotteries with their complete prize structures. I implemented ticket verification through QR code scanning and manual number entry, and automated result scraping scheduled to match the official draw times. I also added Sinhala-to-romanized-English name conversion for localized display, PWA support, and automated winner email notifications. To manage time-series result data efficiently, I used QuestDB and containerized the system with Docker Compose.

Technologies: Next.js frontend, Express.js backend, QuestDB, Docker Compose, PWA, web scraping automation.

GitHub: Lotto-scan

Aya – AI-Powered Menstrual Health & Cycle Tracking Application *Two-Person Team Project / Ongoing*

In this project, I am focusing on the backend and data layer of a menstrual health application. I am developing RESTful services with Node.js and Express.js for cycle tracking, symptom logging, user profiles, and personalized insights. I designed the Supabase PostgreSQL database with relational schemas, validation, and Row-Level Security (RLS) policies to keep each user's health data private. I am also building a cycle prediction algorithm that supports regular and irregular cycles, using average cycle length, variability, and standard deviation to produce predicted ranges with confidence levels. In addition, I integrated the Claude API to power a context-aware, non-diagnostic chatbot with personalized responses and red-flag symptom detection, and I collaborate with my teammate on API contracts, frontend integration, and cloud deployment.

Technologies: Node.js, Express.js, PostgreSQL with Supabase, Claude API, REST APIs, Row-Level Security.

GitHub: aya

Small Shop Inventory + Reorder Alert System *Personal Project*

In this project, I built a full-stack inventory management platform for small shop owners to replace manual stock tracking. I implemented barcode scanning through the phone camera for scan-to-sell workflows, automatic stock decrementing on each sale, and low-stock reorder alerts. I designed a relational database schema with PostgreSQL (Supabase) and Prisma ORM to keep stock and sales records consistent, and structured it to support future additions such as stock movement logs, supplier price history, and batch/expiry tracking. I also implemented role-based access control for owners and staff, and a sales dashboard that shows best- and worst-selling items.

Technologies: Next.js 14, TypeScript frontend, Node.js/Express.js backend, PostgreSQL with Supabase, Prisma ORM.

GitHub: shop-inventory-system

Indoor Air Quality Monitoring System *Team Project*

In this project, our team built a real-time indoor air quality monitoring system around the ESP32-WROOM-32. The device measures particulate matter (PM1.0/2.5/10), VOCs/NOx, O_3, CO_2, temperature, and humidity through a multi-sensor array (PMS5003, SGP41, ZMOD4510, MH-Z19, DHT22, DS18B20) and calculates a real-time AQI based on Sri Lankan air quality standards. To keep data reliable during connectivity outages, the system uses local flash storage with RTC-timestamped logging and dual Wi-Fi/Ethernet redundancy for cloud sync to Firebase. We developed a React and Vite web dashboard for real-time and historical data visualization, and the team also designed a custom PCB in EasyEDA and a 3D-modeled enclosure in Blender. The project was presented at FIT EXPO 2025.

Technologies: ESP32, C/C++ (Arduino IDE), Firebase Realtime Database, React, EasyEDA, Blender.

Contributions

Road to Legacy 2.0 *Delegate Handling Committee Member*

Radiance 2.0 *Finance Committee Member*
IEEE Student Branch, University of Moratuwa

Courses

Data Science for Beginners
IBM

Python For Data Science
IBM

Web Design for Beginners
Centre for Open and Distance Learning, University of Moratuwa

Python for Beginners
Centre for Open and Distance Learning, University of Moratuwa (Progressing)

References

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